



Fusion of ECA-Net with Hamiltonian Deep Neural Networks and Dynamic Step Size Optimization

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Abstract. The Hamiltonian Deep Neural Networks (H-DNNs) is an innovative neural network architecture that integrates the energy conservation and gradient stability principles from Hamiltonian mechanics, effectively addressing the gradient vanishing and interpretability issues commonly encountered in traditional deep learning methods. Firstly, this study empirically validates the effectiveness of H-DNNs in complex tasks, demonstrating its applicability to challenging image recognition problems. However, the H-DNNs employs a fixed time step, which presents limitations when handling complex data and fails to adequately capture critical features during the model training process. To address these issues, a novel architecture, Adaptive Hamiltonian Deep Neural Networks combined with Efficient Channel Attention (AdaHamil-ECA), is proposed in this paper. This architecture introduces an adaptive time-step mechanism that intelligently adjusts the time step based on data complexity and the model's learning state, ensuring that the model has sufficient time to integrate and propagate information when processing complex features while avoiding unnecessary computational resource usage. Additionally, the embedded ECA-Net module in AdaHamil-ECA learns the importance and interrelationships of different channels, precisely enhancing the weights of critical feature channels while suppressing irrelevant ones. This enables the model to focus more on the feature information that significantly contributes to the task goal, thereby improving discriminative power and generalization performance. Experimental results show that AdaHamil-ECA outperforms traditional H-DNNs by a 2.7%-3% increase in accuracy for complex image classification tasks, while also accelerating convergence and significantly enhancing robustness.

Keywords: Hamiltonian Deep Neural Networks · Adaptive time-step mechanisms · ECA-Net · Natural Language Processing · Gradient stability

1 Introduction

In recent years, H-DNNs have emerged as a novel neural network architecture, attracting widespread attention in the field of deep learning [1]. The core advantage of H-DNNs lies in their ability to address inherent issues present in traditional deep learning models. During the training process of conventional deep learning models, problems such as vanishing or exploding gradients often arise in deep networks, which limit the depth and performance of the network. By incorporating parameterized Hamiltonian systems and symplectic geometric integration algorithms, H-DNNs ensure the stability of gradients throughout the training process, effectively overcoming these challenges. Furthermore, H-DNNs offer improved interpretability, with their design inspired by Hamiltonian mechanics. Each layer of the network corresponds to energy transformation and conservation processes within a physical system, thus enhancing the transparency of the internal mechanisms of the network. Researchers can analyze the relationship between network parameters and the Hamiltonian function to observe energy changes during the training process, thereby gaining deeper insights into the model's decision-making process and feature learning mechanisms. Due to their gradient stability and interpretability, H-DNNs have demonstrated significant potential in deep feature extraction tasks, particularly in fields such as natural language processing and image processing. Not only do H-DNNs provide a new theoretical framework for deep learning, but they also offer crucial support for model optimization and deployment in practical applications.

2 Related Work

The integration of Hamiltonian mechanics with neural networks applies the principles of Hamiltonian dynamics to deep learning, addressing issues such as vanishing gradients and the lack of interpretability in deep neural networks. Clara Lucía Galimberti et al. [2] proposed a deep neural network framework based on the time discretization of Hamiltonian systems (H-DNNs) and demonstrated its practicality in MNIST handwritten digit recognition [3]. Initial explorations have also been made in the field of natural language processing. Additionally, the CIFAR-10 dataset [4] was mentioned in the appendix, and experiments were conducted, showing that the classification accuracy of H-DNNs was only slightly behind the popular ResNet model (The dataset is shown in Table 1). This suggests that fine-tuning the network model could enhance H-DNNs and lead to more widespread applications. Allen-Blanchette et al. [5] combined generative adversarial networks with Hamiltonian mechanics, proposing the Hamiltonian Generative Adversarial Network (HGAN) model for video generation based on physical laws. Narahari Ajmeera et al. [6] introduced an emotion analysis model that combines Hamiltonian mechanics with deep neural networks, advancing the field of natural language processing. Marin et al. [7] explored the enhancement of artificial intelligence reasoning abilities using Hamiltonian mechanics, aiming to improve multi-hop reasoning capabilities within AI systems. These contributions indicate that research on Hamiltonian mechanics continues to progress.

Inspired by the work of Clara Lucía Galimberti et al., this study explores the potential applications of the H-DNNs model in natural language processing. However, limitations

were identified when dealing with complex tasks. To address these issues, this paper proposes an improved model, AdaHamil-ECA, which introduces an adaptive time-step mechanism to enhance training efficiency and numerical stability, while incorporating a lightweight ECA-Net module to optimize feature propagation. These improvements effectively overcome the performance bottlenecks of H-DNNs, providing new directions for future development. Since the application of H-DNNs in natural language processing is still in its early stages, challenges such as training efficiency bottlenecks and insufficient feature representation are evident when dealing with complex datasets and tasks. Therefore, this paper aims to improve the model's performance by thoroughly investigating its applicability and making targeted enhancements, thus laying a foundation for future research and applications.

Table 1. Dataset Introduction

Dataset Name	Image Size	Image Type	Number of Categories	Dataset Size	Dataset Features
MNIST	28×28 pixels	Grayscale Images	10 categories	Training set: 60,000 images; Test set: 10,000 images	Standard dataset, clearly displays numbers 0 to 9, images have been standardized and normalized
CIFAR-10	32×32 pixels	Color Images	10 categories	Training set: 50,000 images; Test set: 10,000 images	Color images, rich in categories, high image complexity, suitable for evaluating model performance in color image tasks

3 Hamiltonian Deep Neural Network

This section of the study will conduct a series of comparative experiments to thoroughly verify the applicability of the H-DNNs model in image classification tasks.

3.1 Experimental Setup

This section systematically investigates the generalization ability of the H-DNN model by expanding the task from 10 handwritten digit categories to 37 English letter categories, and then to 3,755 Chinese character categories. The experimental design spans a range of tasks, from simple handwritten digits to complex Chinese character images, evaluating the model's performance across multiple levels of difficulty. This comprehensive assessment provides empirical evidence for the model's broad applicability. To achieve this, two representative public datasets were selected for this study: the EMNIST dataset [8] and the HWDB1.1 dataset [9]. Detailed information about these datasets is provided in Table 2.

Table 2. Dataset

Dataset Name	Image Type	Categories	Dataset Size	Dataset Features
EMNIST	Grayscale Images	37 categories	Training set: 88,800 images; Test set: 14,800 images	An extended version of the MNIST dataset, designed to facilitate research and comparison under the same model architecture
HWDB1.1	Grayscale Images	3755 categories	Training set: 200,000 images; Test set: 40,000 images	Comprising 300 individuals' handwritten characters, this dataset features a wide variety of character types and high image complexity, making it suitable for evaluating model performance

3.2 Experimental Results

To compare the performance of H-DNNs with existing methods, experiments were conducted on the EMNIST and HWDB1.1 datasets. The experimental results are presented in Table 3.

1. On the EMNIST dataset, the following models were selected for comparison: CNN (flat; 2 conv+1 dense) [10]; Committee of 7 CNNs [11]; Record-to-Record Travel [12]; Hidden Markov Model [13] and Spiking Neural Network using SpykeFlow [14].

Table 3. Comparison of Experimental Results

Dataset	Domain	Categories	Model	ACC (%)
EMNIST	Handwritten English Characters	37categories	CNN(flat; 2conv+1dense)	93.63
			Committee of 7 CNNs	92.42
			Record-to-record travel	93.78
			Hidden Markov model	90
			Spiking Neural Network using SpykeFlow	94.49
			H-DNNs	94.56
HWDB1.1	Handwritten Chinese Characters	3755categories	ResNet	93.30
			CBAM-ResNet	94.41
			ViT	93.85
			CNN-Median Filtering	92.28
			H-DNNs	93.53

2. On the HWDB1.1 dataset, the following models were chosen: ResNet [15];CBAM-ResNet [16];ViT [17] and CNN with Median Filtering [18].

The experimental results demonstrate that the H-DNNs model exhibits remarkable versatility in the field of handwritten character recognition. First, on the EMNIST dataset, H-DNNs achieved an accuracy of 94.56%, surpassing traditional methods such as CNNs and Spiking Neural Networks, thereby validating its superiority in moderately complex tasks. Furthermore, when tested on the HWDB1.1 dataset, a high-complexity scenario, H-DNNs still attained an accuracy of 93.53%. Although slightly lower than CBAM-ResNet (94.41%), it significantly outperformed mainstream models like ResNet (93.30%) and CNN (92.28%), indicating its competitiveness in large-scale classification tasks. The following conclusions can be drawn:

1. Feasibility for practical applications: The model maintains high performance in large-scale tasks (e.g. 3755 categories), demonstrating excellent scalability and potential for broad application in tasks of varying sizes and real-world scenarios.
2. Adaptability to data diversity: Despite the increase in data complexity—from simple English letters to intricate Chinese characters—model performance remained unaffected, suggesting its capacity to handle diverse data types effectively.

In conclusion, experiments show that the H-DNNs network model has significant potential. It provides a solid theoretical framework with good interpretability, offering a logical basis for complex problems. In practical training tasks, H-DNNs, with its unique architecture and optimization strategies, demonstrates excellent performance and the ability to quickly adapt to various data types and task requirements, driving technological development and breakthroughs.

The experiments reveal certain limitations of the H-DNNs model. Specifically, the fixed time step in the Hamiltonian layer cannot be adjusted dynamically based on the complexity of the input data (e.g. text images with varying resolutions), which restricts the model's adaptability. Additionally, H-DNNs lack a hierarchical attention mechanism, making it difficult to accurately assign weights to key region features when processing complex glyph structures (e.g. handwritten characters). This limitation prevents effective modeling of fine-grained spatial correlations, which in turn affects the distribution of feature weights and leads to a decline in performance for complex tasks. These constraints may pose significant bottlenecks to the development of H-DNNs in high-precision and highly adaptive scenarios.

4 Translation of the Improved Model

4.1 Overall Design

This section presents the AdaHamil-ECA framework, which overcomes the limitations of traditional H-DNNs in dynamic adaptability and feature representation. (as shown in Fig. 1). The framework integrates an adaptive time step mechanism with a lightweight ECA-Net. The former dynamically adjusts the time step based on the complexity of the input data, addressing the limitations of fixed time steps that hinder adaptability. The latter employs a channel attention module to precisely weight key features, optimizing feature representation for complex glyph structures. This combination enhances the model's flexibility and efficiency, significantly improving performance on complex tasks.

In traditional H-DNNs, the fixed time step limits training flexibility, resulting in slow convergence and difficulty adapting to complex dynamic systems. To address this issue, this study proposes an adaptive time step algorithm based on gradient variance. The algorithm dynamically adjusts the integration step size by monitoring the stability of parameter updates: the step size is increased when the error is small to improve efficiency, and reduced when the error is large to ensure accuracy. Additionally, a time step variation factor boundary is introduced to optimize the adjustment range and prevent numerical instability. This adaptive algorithm significantly enhances the model's convergence speed, dynamic adaptability, and feature representation ability, ensuring efficient learning in complex tasks. Furthermore, to improve feature representation, a lightweight ECA-Net module is embedded in the feature transmission path. ECA-Net employs an efficient channel attention mechanism, using 1D convolution operations to selectively filter key features and optimize the flow of information. This avoids the dimensionality reduction and expansion processes of traditional SE modules, not adding significant computational burden. Its core design is the adaptive convolution kernel size, which can be dynamically adjusted based on the channel dimension, allowing for flexible capture of local cross-channel interaction information. This lightweight mechanism strengthens the model's focus on important features, enhancing its ability to distinguish features and giving it a competitive advantage in complex image tasks.

In the experiments, the AdaHamil-ECA framework, by embedding the lightweight ECA-Net module and introducing an adaptive time step mechanism, significantly

enhanced model performance. After the ECA-Net module is inserted into the convolutional layer, it applies 1D convolution to the globally average-pooled features, performing weighted processing and dynamically adjusting channel weights to accurately select key features. This module adaptively adjusts the convolution kernel size, optimizing cross-channel interactions, reducing computational overhead, and improving both feature extraction efficiency and discriminative ability. Furthermore, within the Hamiltonian layer, the adaptive time step mechanism dynamically adjusts the integration step size, substantially accelerating the convergence speed and improving the numerical stability of the training process. Specifically, the model introduces a learnable logarithmic step size parameter, combined with a decay factor, to dynamically adjust the time step at each layer. This design enables the model to flexibly adjust the step size based on the complexity of the input data and the progress of training, thereby achieving greater flexibility and efficiency during the training process.

In summary, the AdaHamil-ECA framework significantly enhances feature capture ability and training efficiency through the embedding of the ECA-Net module and the adaptive time step mechanism. The ECA-Net module optimizes the feature propagation path, strengthening the focus on key features, while the adaptive time step mechanism dynamically adjusts the step size, accelerating convergence and ensuring numerical stability. This provides strong support for efficient learning in H-DNNs and their application to a variety of tasks.

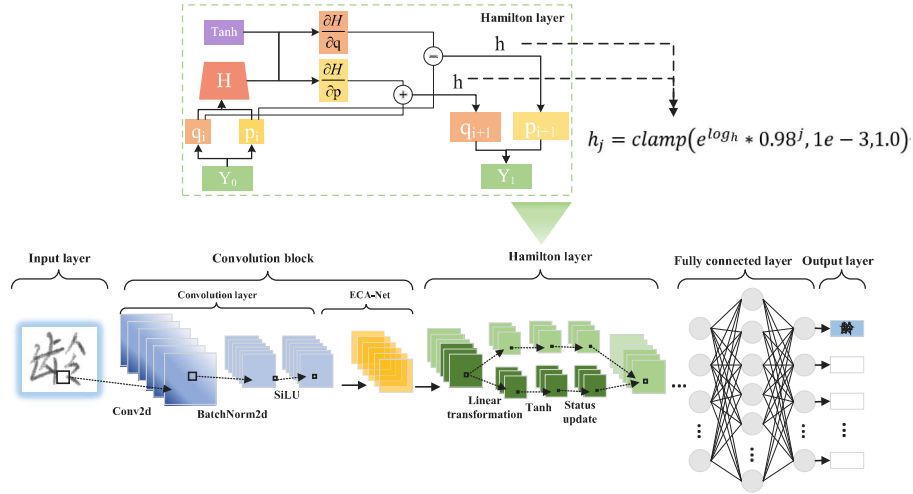


Fig. 1. AdaHamil-ECA Model Diagram

4.2 Adaptive Time-Step

The adaptive time-step is a dynamic adjustment strategy that flexibly alters the integration step size based on the real-time state of the system. It is widely used in numerical simulations and the analysis of complex dynamic systems. Unlike traditional numerical

integration methods, which typically use a fixed time-step, the latter often struggles to maintain computational accuracy when faced with rapid changes in the system's state and is unable to adapt to varying levels of input data complexity. In contrast, the adaptive time-step monitors the rate of change in the system's state in real time and automatically adjusts the step size, thereby significantly enhancing computational efficiency while ensuring accuracy.

In this study, an adaptive time-step mechanism is implemented, combined with an inter-layer step decay strategy and parameter space constraints to preserve the local symplectic structure of the symplectic Euler discretization. This design not only enhances the model's adaptability when handling complex dynamic systems but also effectively balances numerical stability with computational efficiency. Based on this, Eq. (1) further details the implementation of this mechanism. Through this approach, the study provides a solid theoretical foundation for the engineering applications of H-DNNs, ensuring both the efficiency and accuracy of the computational process.

$$h_j = \text{clamp}\left(e^{\log h} * 0.98^j, 1e-3, 1.0\right) \quad (1)$$

In this context, h_j represents the time step for the j -th layer, while $e^{\log h}$ denotes the base time step, with the exponential function ensuring its positivity. The factor 0.98^j serves as a decay factor, causing the time step to decrease progressively with the increase in network layers. This design allows for the use of larger time steps in the shallow layers, enabling the rapid capture of global features, while smaller time steps in the deeper layers facilitate more precise adjustments to local features. The clamp function restricts the time step within the range of $[1e-3, 1.0]$, preventing it from becoming either too large or too small.

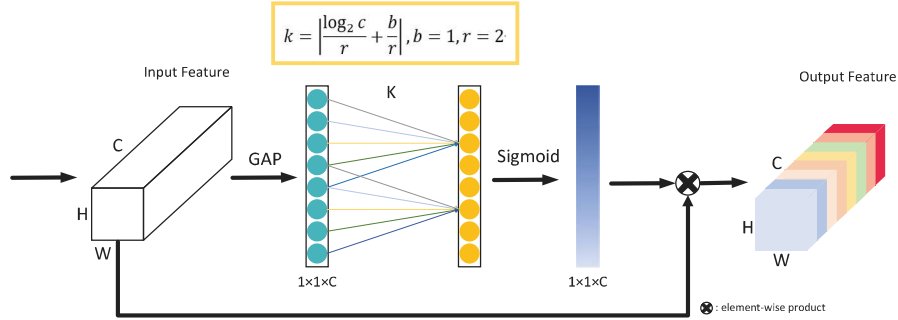


Fig. 2. ECA-Net

4.3 ECA-Net

In the task of Bengali handwritten compound character recognition, characters are composed of multiple basic characters, exhibiting complex shapes and diverse writing styles,

which presents significant challenges for recognition. To enhance recognition performance, Mohammad Meraj Khan et al. [19] introduced the SE module-based ResNeXt deep convolutional neural network model. However, SE-Net is constrained in practical applications due to its high parameter count and computational complexity. In contrast, ECA-Net captures channel dependencies by adaptively adjusting the convolution kernel size, offering a lower parameter count, reduced computational complexity, higher training efficiency, and better adaptability.

ECA-Net [20] is an efficient channel attention mechanism that captures global inter-channel dependencies through 1D convolution and dynamically assigns weights to each channel (as shown in Fig. 2). Compared to traditional attention mechanisms, ECA-Net can automatically learn the importance of channels, enhancing the model's focus on critical features, improving feature representation and discrimination capabilities, while being more computationally efficient and requiring fewer parameters. By integrating ECA-Net into H-DNNs, the deficiencies in feature representation and adaptability of H-DNNs can be effectively addressed. Although H-DNNs perform excellently in gradient stability, their feature extraction ability still requires further enhancement when dealing with complex features and diverse tasks. The introduction of ECA-Net enables H-DNNs to more precisely capture inter-channel correlations and automatically focus on the key features of the task, significantly improving the model's performance across various tasks and datasets.

Furthermore, ECA-Net has a lower computational complexity, ensuring that these improvements do not significantly increase computational costs, thus making H-DNNs more practical in resource-constrained environments. Additionally, by dynamically adjusting channel weights, ECA-Net further enhances the model's generalization ability, resulting in superior performance on diverse datasets and thus improving the adaptability and performance of H-DNNs in practical applications.

4.4 Experimental Results

This section systematically evaluates the performance improvements of the proposed AdaHamil-ECA model through ablation experiments. Four representative public datasets were selected for the experiments: MNIST, EMNIST, HWDB1.1, and CIFAR-10, covering a range of diverse task scenarios, from simple handwritten digits to complex Chinese characters and natural images. The results are presented in Table 4. The experiments compared four models: the baseline H-DNNs, Ham-h with adaptive time steps, Ham-ECA with the embedded ECA-Net mechanism, and the complete improved framework, AdaHamil-ECA.

Table 4 presents the experimental results, demonstrating that the AdaHamil-ECA model achieves significant performance improvements over the baseline H-DNNs across three public datasets, with accuracy increases of 2.69%, 2.03%, and 4.32%, respectively. This substantial performance enhancement provides strong evidence of the critical role played by the adaptive time step mechanism and the ECA-Net module in enhancing model effectiveness. The adaptive time step mechanism significantly accelerates convergence and improves numerical stability during training by dynamically adjusting the

Table 4. Ablation Experiment Results

Dataset	Categories	Model	ACC (%)
EMNIST dataset	37 categories, handwritten English characters, grayscale images	H-DNNs	94.56
		Ham-h	96.56
		Ham-ECA	95.43
		AdaHamil-ECA	97.25
HWDB1.1 dataset	3755 categories, handwritten Chinese characters, grayscale images	H-DNNs	93.53
		Ham-h	96.11
		Ham-ECA	95.47
		AdaHamil-ECA	97.85
CIFAR-10 dataset	10 categories, common objects such as airplanes, cats, and birds, color images	H-DNNs	92.27[2]
		Ham-h	93.4
		Ham-ECA	92.98
		AdaHamil-ECA	94.3

integration step size, thereby providing robust support for efficient model training. Meanwhile, the ECA-Net module employs an efficient channel attention mechanism to precisely select key features, optimize the feature propagation path, and further strengthen the model’s ability to express features in complex image tasks. This improvement enables the model to more accurately capture essential information in images, thereby boosting its overall performance.

When only the adaptive time step mechanism (Ham-h) was introduced, the model’s accuracy improved by 2%, 2.58%, and 1.13% on the three datasets, respectively. In contrast, when only the ECA-Net module (Ham-ECA) was introduced, the accuracy increased by 0.87%, 1.49%, and 0.71%, respectively. These results further highlight the significant contributions of both mechanisms to the model’s performance: the adaptive time step mechanism enhances the model’s dynamic adaptability by optimizing the training process, while the ECA-Net module improves performance by strengthening feature expression capabilities. Together, these mechanisms complement each other, driving the exceptional performance of the AdaHamil-ECA model in complex tasks.

The results from the comprehensive ablation study demonstrate that the AdaHamil-ECA model significantly improves training efficiency and feature expression ability through the adaptive time step mechanism and the ECA-Net module. The adaptive time step mechanism optimizes the training process, ensuring both efficiency and stability, while the ECA-Net module, through its channel attention mechanism, precisely selects key features, thereby enhancing the model’s ability to understand complex tasks. The two mechanisms work synergistically to lay the foundation for the model’s high performance, with the removal of either mechanism leading to a decline in performance. This improvement provides strong support for the application of H-DNNs in complex tasks and offers valuable insights for future research.

5 Conclusion

This study investigates the performance of H-DNNs in complex image tasks and introduces the improved AdaHamil-ECA model. The model enhances its adaptability to complex tasks by incorporating an adaptive time step mechanism that dynamically adjusts the time step. Additionally, the ECA-Net module is embedded, leveraging channel attention mechanisms to strengthen the representation of key features. Experimental results show that AdaHamil-ECA achieves an average improvement of 2.7% to 3% in classification accuracy compared to the baseline H-DNNs, demonstrating its superiority. In the future, the application of AdaHamil-ECA can be expanded to more complex multimodal tasks, such as video analysis and 3D image recognition, and further performance enhancements can be achieved by integrating cutting-edge technologies like Transformers and Graph Neural Networks (GNNs). This opens up new possibilities for the application of deep learning in a wider range of fields.

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